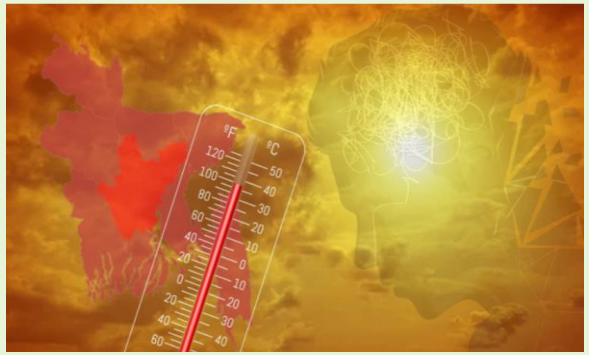
Predicting Thermal Comfort Impacts of Urban Water Shrinkage in Dhaka Using Artificial Learning Models



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Outline

1. Introduction
2. Research Aim and Objectives
3. Study Area
4. Methods
5. Results
6. Conclusions

Introduction: Thermal condition and Waterbody Loss in Highly Populated Areas

- Dhaka's urban areas are experiencing rising temperatures due to the loss of waterbodies and dense population, affecting land surface temperature and thermal comfort.

- These changes threaten human comfort, city infrastructure, and urban ecosystems.
- Dhaka is home to over 20 million people, many affected by rising temperatures and shrinking waterbodies.

- Assessing comfort zones reveals heat stress patterns and guides measures to improve urban thermal comfort.



https://www.dhakatribune.com

Introduction: Impacts of Elevated LST and Waterbody Loss in Dhaka City

Health Risks: Heat stress affects vulnerable populations.

Microclimate Alteration: Shrinking waterbodies affect local climate.

Air Quality Deterioration: Rising temperatures can worsen pollution.

Energy Demand Increase: Higher temperatures raise electricity use for cooling.

Water Pollution: Waterbody loss intensifies urban water pollution.

Heatstroke kills 10 people, **DGHS-2024.**



Lake Drying



Air Pollution



Water Pollution



Introduction: Thermal and water shrinkage mapping

- Adaptive capacity reflects how well communities cope with heat stress and water loss.
- Mapping of land surface temperature.

- In Dhaka, local-scale mapping is suitable for assessing adaptive capacity related to urban heat and water loss.

- Predicted heat and water index maps guide heat risk reduction and resource planning.

- A time-series AI model is used to analyze temperature and waterbody changes over time.

Research Aim and Objectives

Aim:

Develops and applies an AI-based geospatial framework to predict and map land surface temperature zones and waterbody shringkage in Dhaka, Bangladesh.

Objectives:

(1) Extract waterbody and LST data to develop artificial intelligence models.

(2) Predict land surface heat and waterbody loss zones using the trained model.

(3) Analyze temperature and waterbody changes over specific time intervals.

Study Area

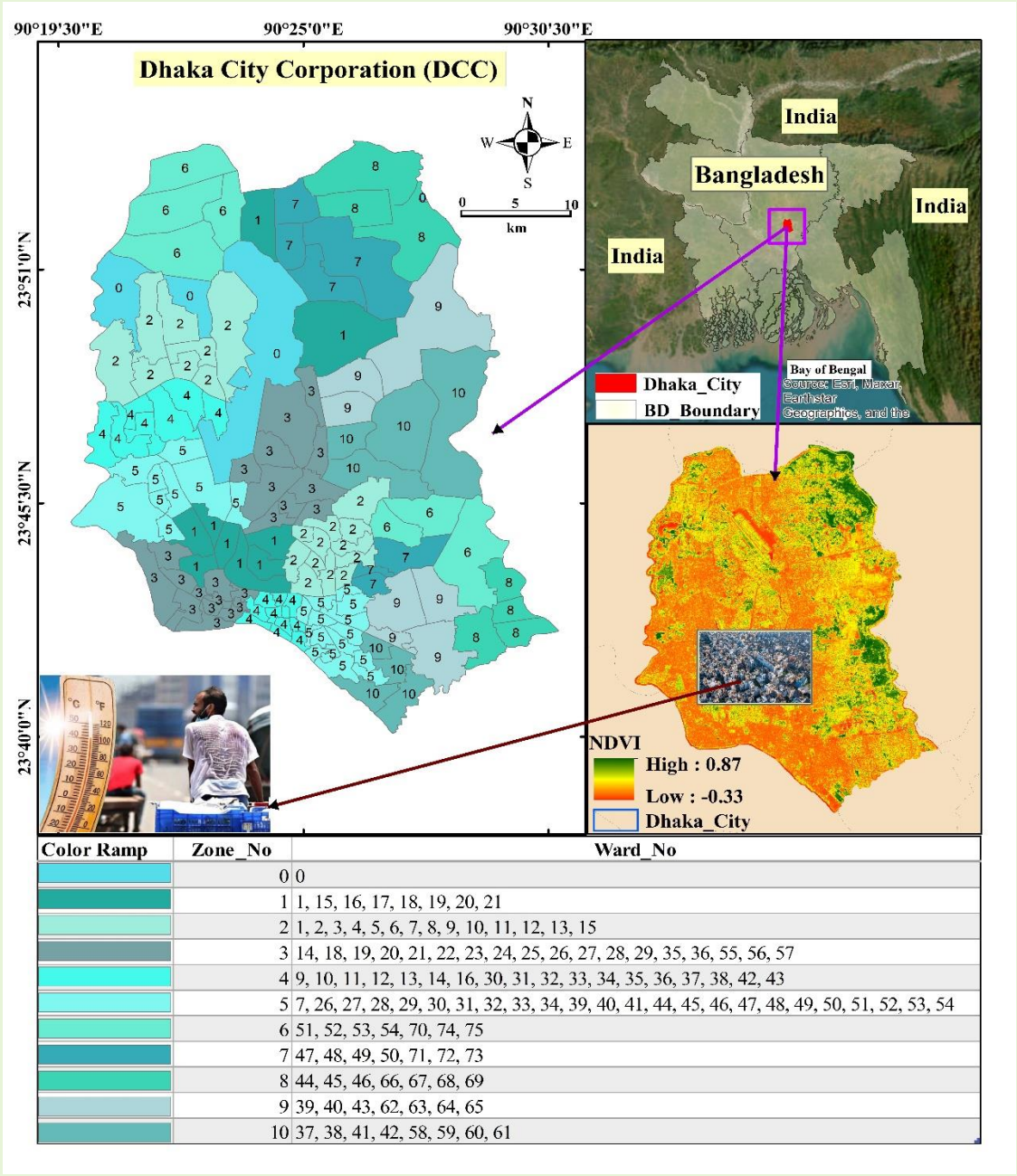
Location: Dhaka Metropolitan City (DMCC), a densely populated urban region in Bangladesh.

Area: approximately 491.9 km².

Population: Around 36.6 million (UN Report, 2025).

Features: Prone to high urban heat, waterbody shrinkage, and related thermal stress.

Location setting: Dhaka, the densely populated economic and political **capital of Bangladesh**, has few remaining natural waterbodies.

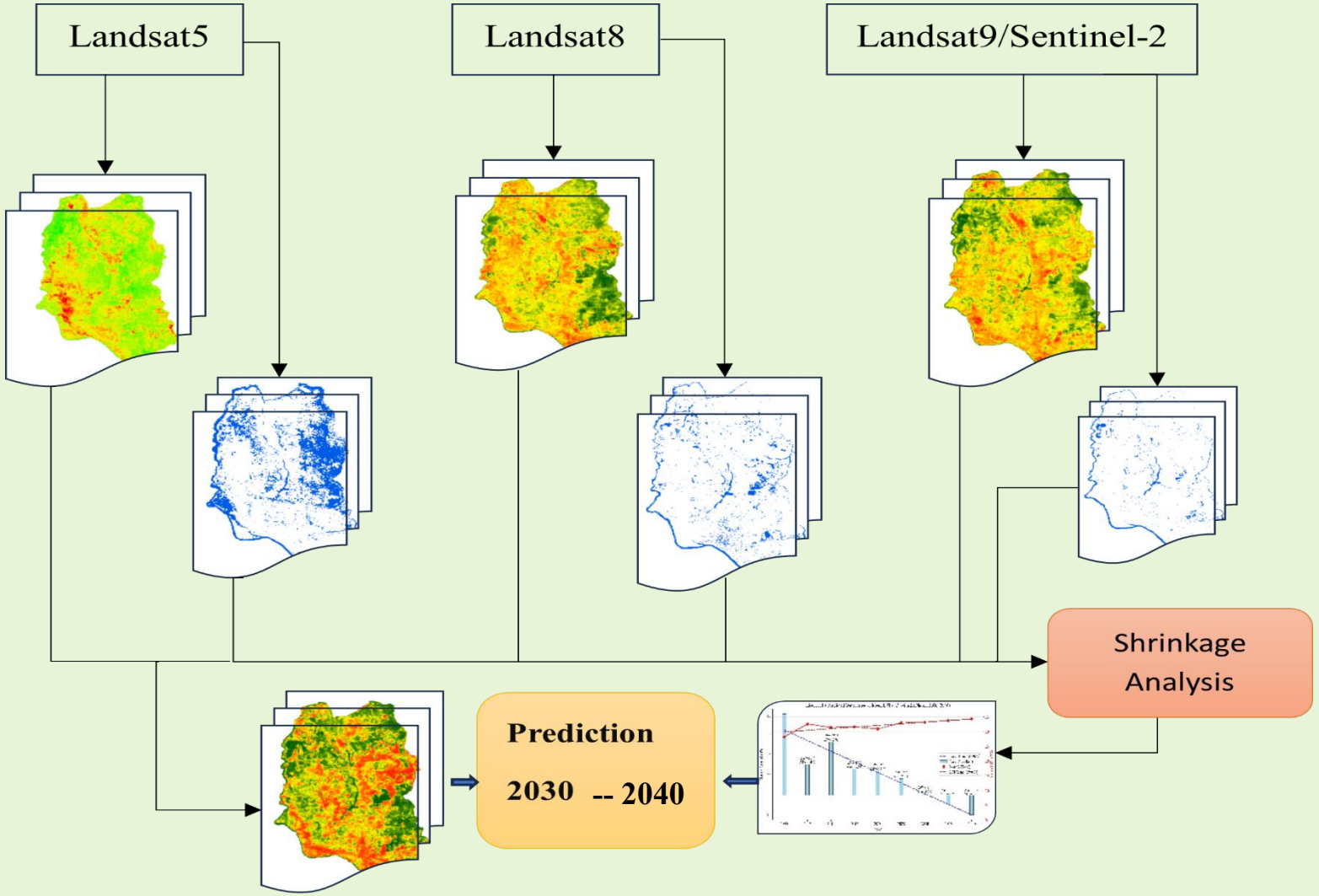


Color Ramp	Zone_No	Ward_No
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	8	44, 45, 46, 66, 67, 68, 69
	9	39, 40, 43, 62, 63, 64, 65
	10	37, 38, 41, 42, 58, 59, 60, 61

Methods: Data source and mapping name

Data Type	Satellite / Sensor	Years	Bands Used	Derived Index / Map	Purpose / Notes
Land Surface Temperature (LST)	Landsat-5/8/9, Sentinel-2	2000–2025	B3, B4, B6	Historical Thermal Maps, Mean LST, ΔLST	Assess urban heat, thermal comfort trends
Waterbody / Wetlands	Landsat/Sentinel	2000–2025	B2, B3, B4, B5, B8	NDWI, MNDWI (> -0.10), NDVI (< 0.3) → Waterbody Maps	Detect surface water extent, monitor shrinkage
Vegetation / Land Cover	Landsat-5/8/9, Sentinel-2	2000–2025	B4, B5, B8	NDVI	Map vegetation cover, greenness trends
Built-up / Urban Expansion	Landsat-5/8/9, Sentinel-2	2000–2025	B3, B4, B5, B6, B11	NDBI (Normalized Built-up), NDTI (Normalized Dryness)	Detect built-up areas and urban growth
Temporal Change Maps	Derived from all above	2000–2025	N/A	ΔLST, ΔWaterbody, ΔVegetation, ΔBuilt-up	Identify trends and hotspot areas over time

Methods: Spatial data processing flow



Methods: Dataset preparing, Temperature conversion and Water detection

- Spatial thematic layers of waterbody and Temperature of each selected year were generated using ArcGIS Pro 3.4 software
- Finalizing projection 46N, extracted target properties.
- Small ponds (NDWI > –0.1) (2) and vegetation (VI < 0.3) were identified using threshold parameters.
- Pixel-based temperature was converted from kelvin (K) to degree Celsius (°C) (3), and NDTI (>0.6) (1) thresholds were used to map heat intensity following these equiation.

$$NDWI=(Green+NIR/Green-NIR)\text{-------}\ (1)\ ; \qquad NDTI=(SWIR+TIR/SWIR-TIR)\ \text{---}\ (2)$$

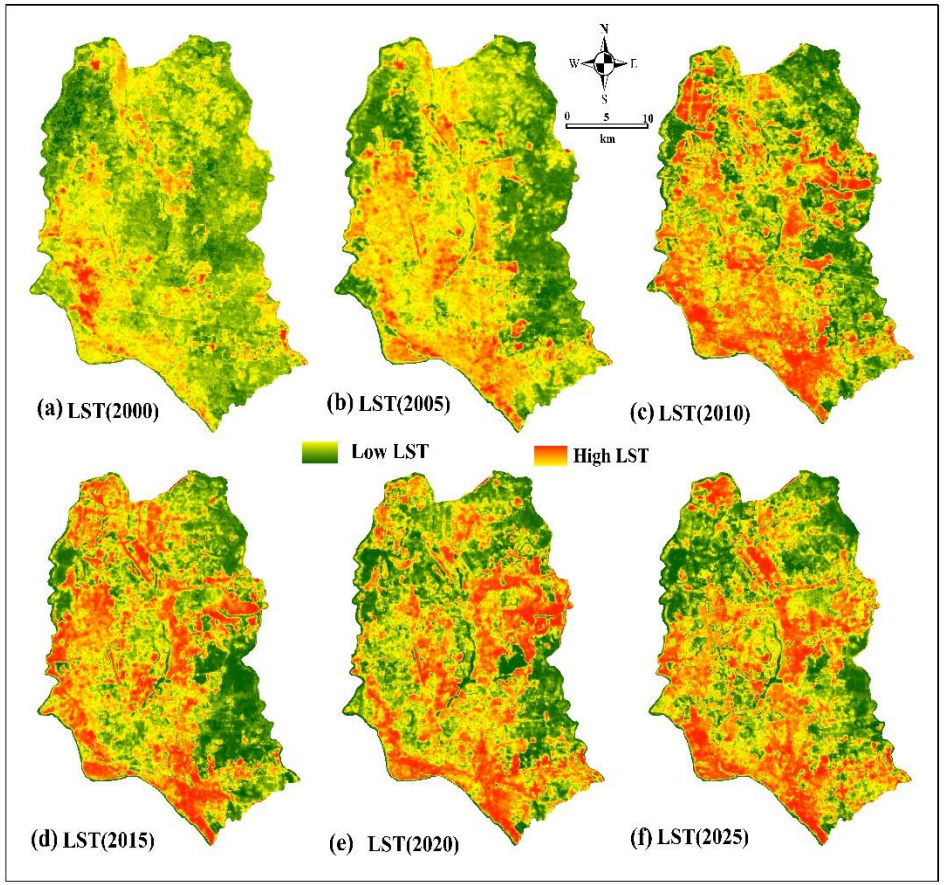
$$T(°C)=T(K)-273.15\ \text{-----}\ (3)$$

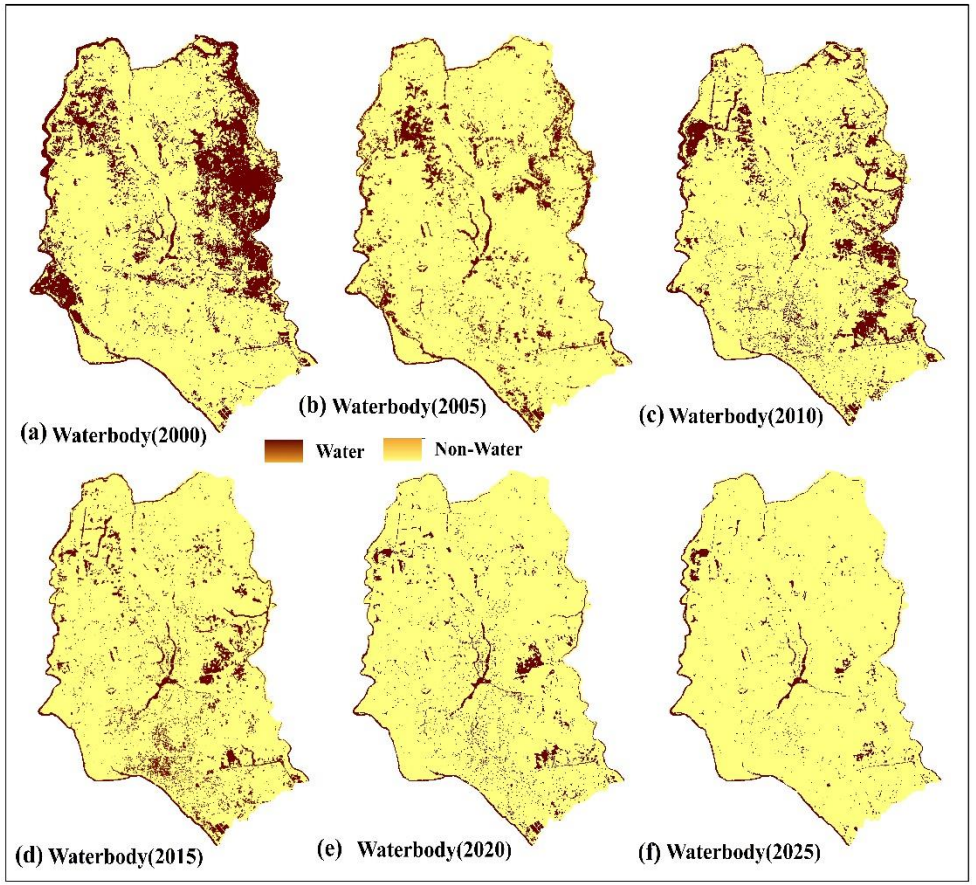
Band Function	Landsat-5 (TM)	Landsat-8/9 (OLI/TIRS)	Sentinel-2 (MSI)
Green	B2	B3	B3
Red	B3	B4	B4
NIR (Near Infrared)	B4	B5	B8
SWIR (Shortwave Infrared)	B5	B6	B11
TIR (Thermal Infrared)	B6	B10/B11	N/A

Methods: Data processing and model use

- Based on IPCC AR6 and literature on urban heat and waterbody changes.
- Data availability and relevance, and their influence on urban heat stress and waterbody changes.
- Datasets were projected, bias-corrected, null values handled, and aligned to a common extent and resolution to minimize errors.
- Deep learning models Temporal Convolutional Network – 1 Dimensional(TCN-1D), Long Short-Term Memory(LSTM)/ Recurrent Neural Network(RNN) were trained using over 3 million points from each raster using 50 epoch and 32 batches.
- Used models on each pixel to predict thermal impact and waterbody changes in Dhaka.

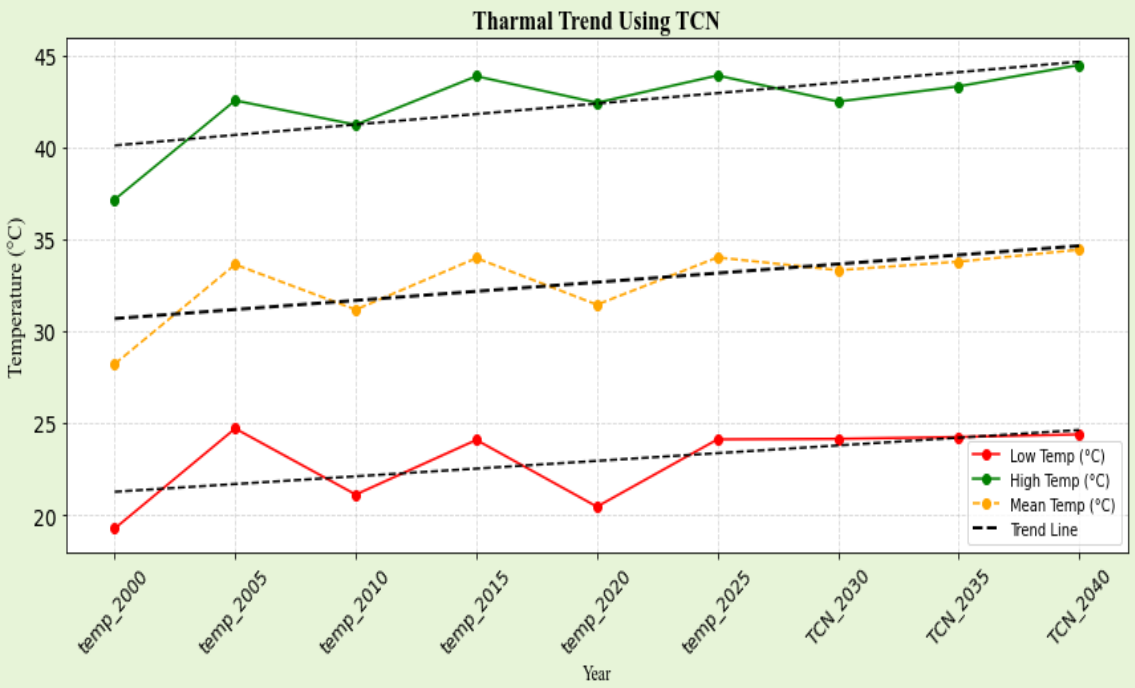
Mapping: Year wise thermal and waterbody visualization





Year	Low (°C)	High (°C)	Mean (°C)	Waterbody (Sq.km)
2000	19.27	37.16	28.22	76.1332
2005	24.71	42.55	33.63	35.1431
2010	21.12	41.23	31.18	50.0151
2015	24.09	43.87	34.98	38.9344
2020	20.46	42.43	31.45	23.3059
2025	24.12	43.91	34.52	17.6318

Result: (Thermal Trend Analysis) Waterbody Shrinkage and thermal change



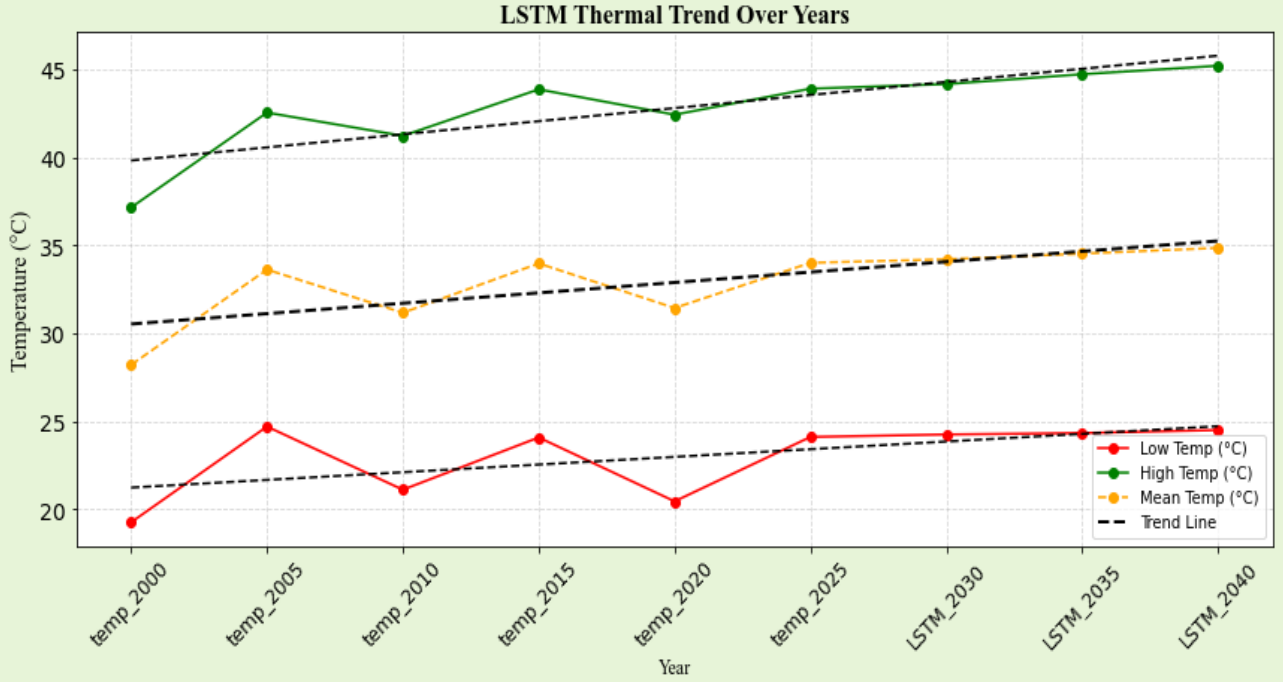
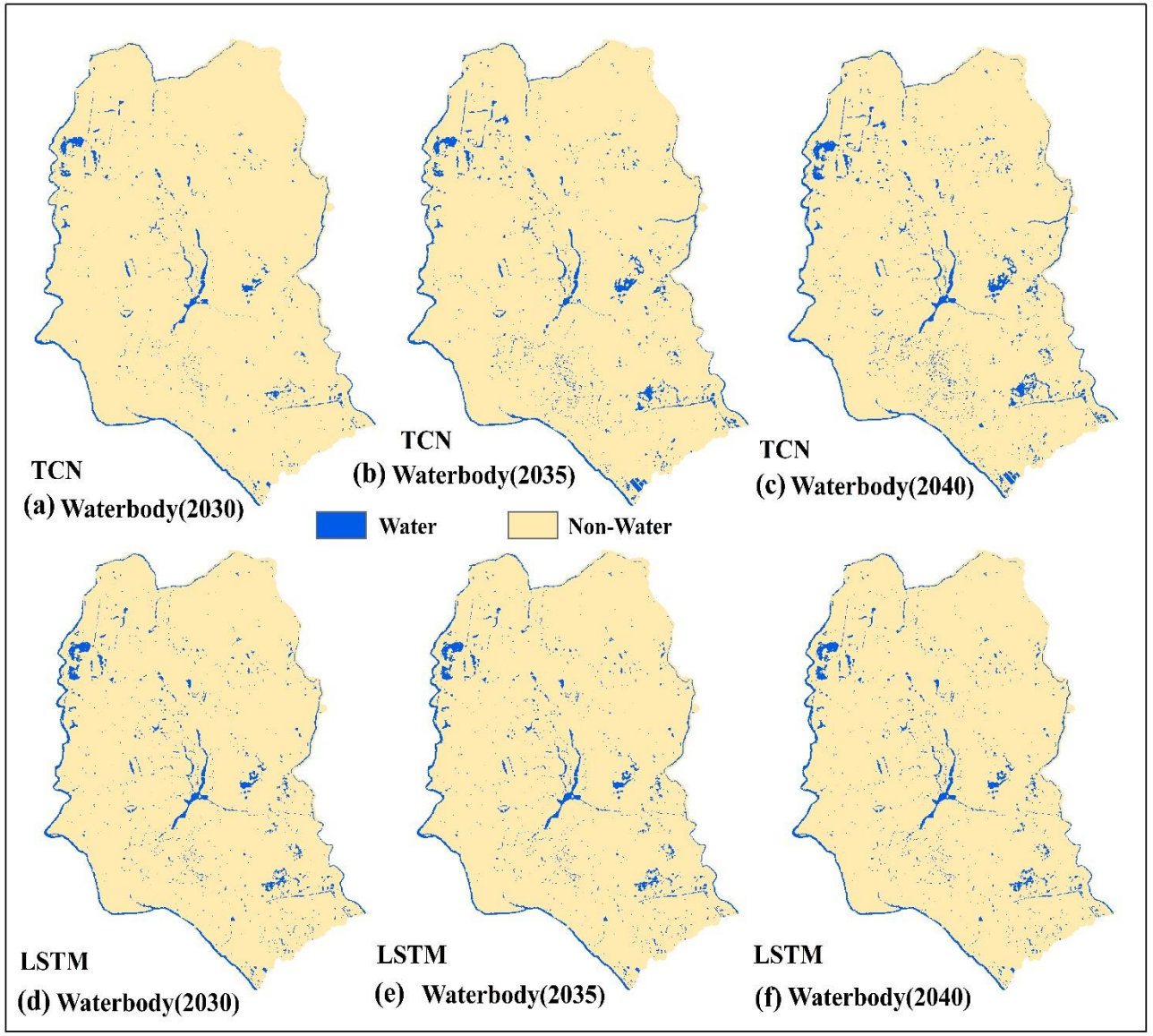


Table: Thermal Change Prediction using Temporal Convolutional Network (TCN) and Long Short-Term Memory (LSTM)

Year (TCN)	Low (°C)	High (°C)	Mean (°C)
2030	24.15	42.49	33.32
2035	24.25	43.32	33.79
2040	24.39	44.48	34.44

Year (LSTM)	Low (°C)	High (°C)	Mean (°C)
LSTM_2030	24.26	44.17	34.22
LSTM_2035	24.34	43.83	34.59
LSTM_2040	24.51	45.73	35.86

Results: Predicted Waterbody Shrinkage



Year (TCN)	Low (°C)	High (°C)	Waterbody(Sq.km)
2030	24.15	42.49	16.1859
2035	24.25	43.32	15.4927
2040	24.39	44.48	15.888

Year (LSTM)	Low (°C)	High (°C)	Waterbody (Sq.km)
2030	24.26	44.17	14.4961
2035	24.34	43.83	14.4354
2040	24.51	45.73	14.2343

Table: Water Change Prediction using Temporal Convolutional Network (TCN) and Long Short-Term Memory (LSTM)

Results: Predicted thermal comfort (TC) zone

- Used time series deep learning model TCN and RNN/LSTM.
- Following 5 years thermal trend models predicted thermal comfort map.
- Here, in the map,
  T= thermal, C= comfortable
  V= very, H= high, L= low,
  M= Moderate.

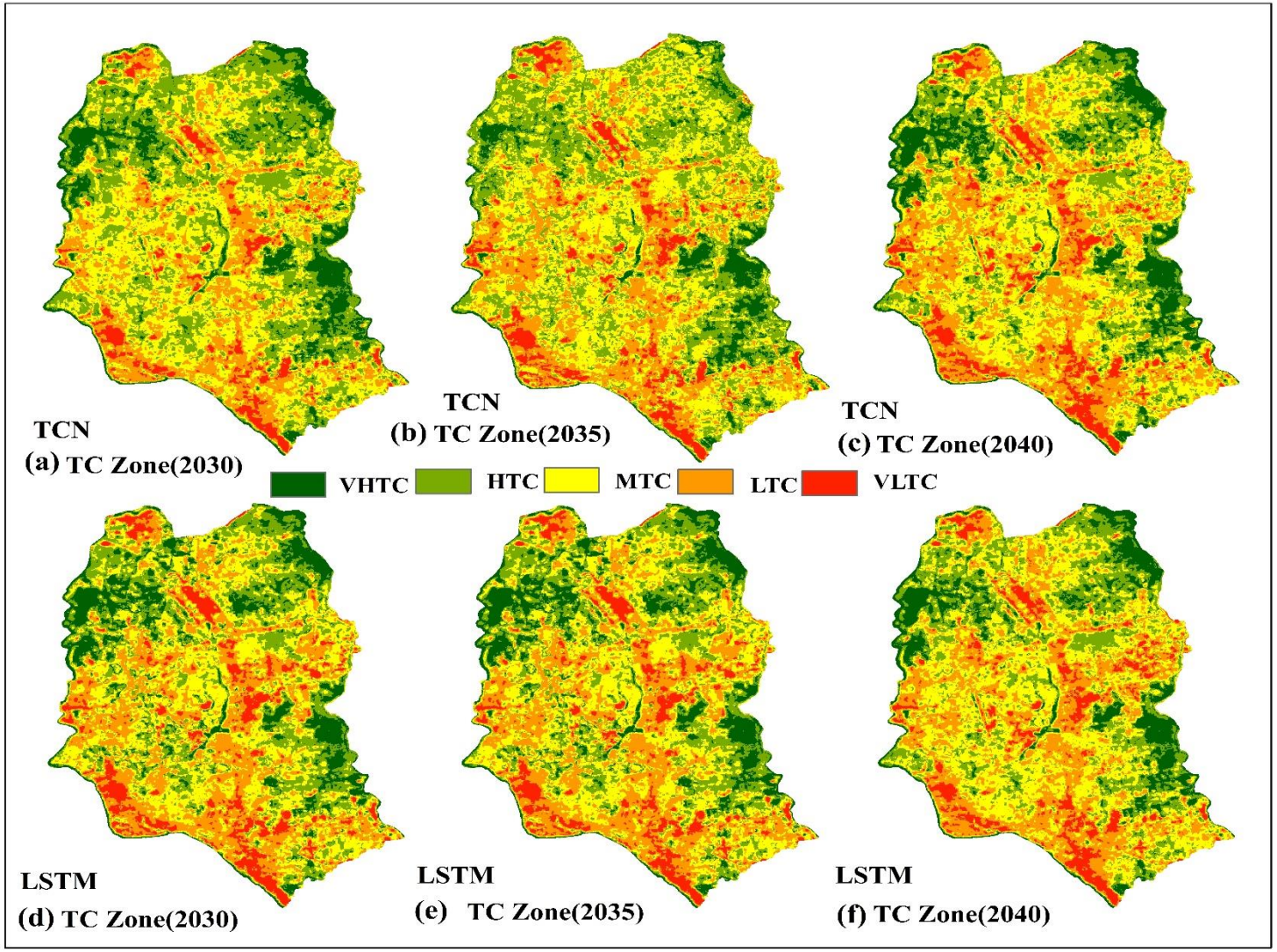
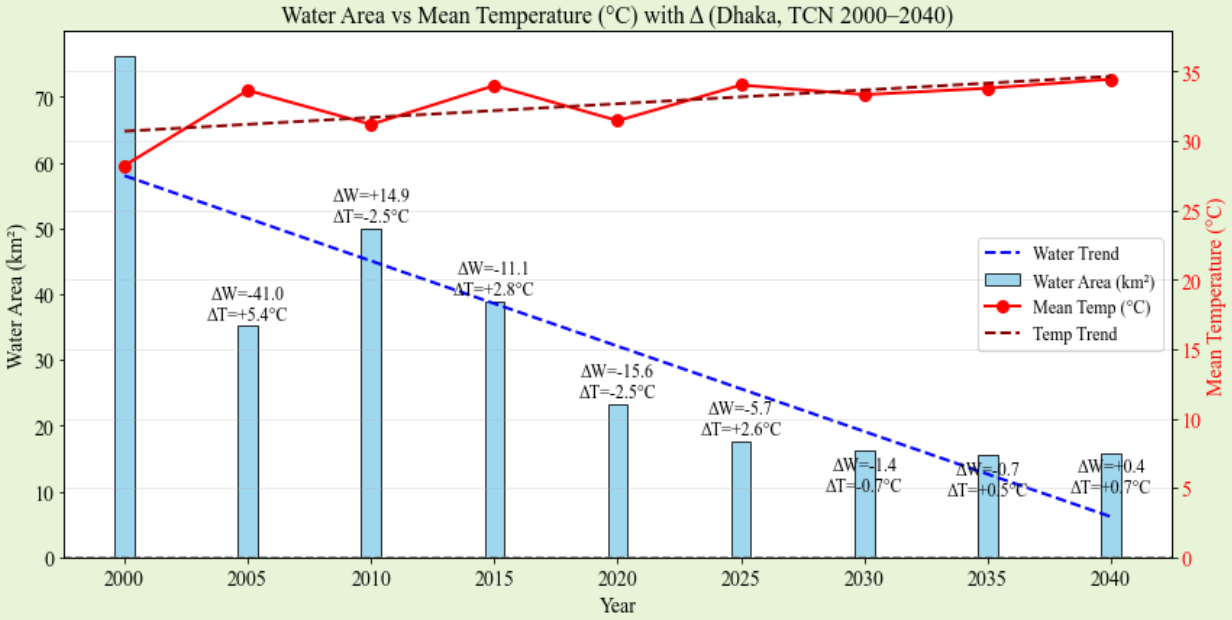


Figure: Thermal comfort map using two different AI models

Results: Temperature Vs waterbody analysis (LSTM Based)

Period	Δ Mean Temp (°C)	Temp Trend	Δ Waterbody (km²)	Waterbody Trend
2000 → 2005	5.41	▲ Increase	−40.99	▼ Decrease
2005 → 2010	−2.45	▼ Decrease	14.87	▲ Increase
2010 → 2015	3.8	▲ Increase	−11.08	▼ Decrease
2015 → 2020	−3.53	▼ Decrease	−15.63	▼ Decrease
2020 → 2025	3.07	▲ Increase	−5.67	▼ Decrease
2025 → 2030	−0.30	▼ Slight Decrease	−3.14	▼ Decrease
2030 → 2035	0.37	▲ Increase	−0.06	▼ Nearly Stable
2035 → 2040	1.27	▲ Increase	−0.20	▼ Decrease

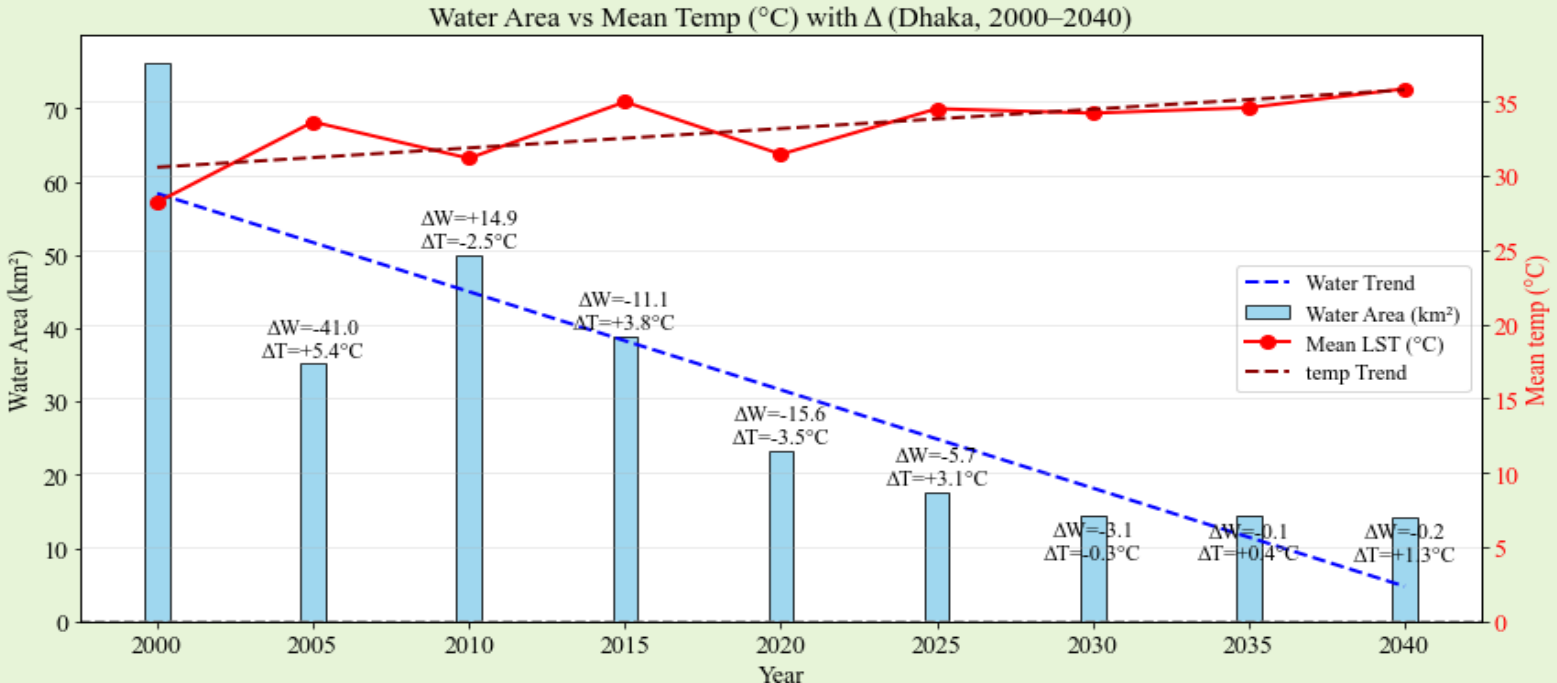


**The Temporal Convolutional Network (TCN) was trained on a dataset of over 3 million samples for 50 epochs with a batch size of 32 and a learning rate of 0.001 using the Adam optimizer. The model achieved moderate prediction accuracy with an $R^2 \approx 0.70$ and a MAPE ≈ 13%, effectively capturing the overall trend of waterbody changes despite some localized variability.

Table: Thermal effect on waterbody prediction using LSTM

Results: Temperature Vs waterbody analysis (TCN Based)

Period	Mean Temp (°C)	Δ Mean Temp (°C)	Temp Trend	Waterbody (km²)	Δ Waterbody (km²)	Waterbody Trend
2000 → 2005	33.63	5.41	▲ Increase	35.14	−40.99	▼ Decrease
2005 → 2010	31.18	−2.45	▼ Decrease	50.02	14.87	▲ Increase
2010 → 2015	33.98	2.8	▲ Increase	38.93	−11.08	▼ Decrease
2015 → 2020	31.45	−2.53	▼ Decrease	23.31	−15.63	▼ Decrease
2020 → 2025	34.02	2.57	▲ Increase	17.63	−5.67	▼ Decrease
2025 → 2030	33.32	−0.70	▼ Slight Decrease	16.19	−1.45	▼ Decrease
2030 → 2035	33.79	0.47	▲ Increase	15.49	−0.69	▼ Decrease
2035 → 2040	34.44	0.65	▲ Increase	15.89	0.4	▲ Slight Increase



**The Long Short-Term Memory (LSTM) model was trained on a dataset of over 3 million samples for 50 epochs with a batch size of 32 and a learning rate of 0.001 using the Adam optimizer. The model achieved moderate prediction accuracy with an $R^2 \approx 0.74$ and a MAPE ≈ 14%, effectively capturing the overall trend of waterbody changes, though some localized variations were not perfectly predicted.

Table: Thermal effect on waterbody prediction using TCN

Conclusions

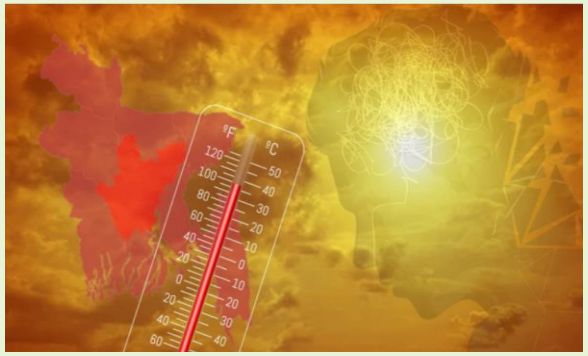
•The analysis of thermal comfort and waterbody shrinkage provides spatial insights for defining urban thermal zones and planning heat mitigation strategies.

•Deep learning-based models (TCN and LSTM) captured long-term trends with moderate accuracy ($R^2 \approx 0.70$-75, MAPE ≈ 13–14%).

•Limitations: Localized variability, driver variables, and future trends may not be fully captured and could be better resolved with a higher volume of data.

Future work will focus on:

- Use higher-resolution data and targeted urban planning to improve predictions and support heat and water management.



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